ARIMA Model for Traffic Flow Prediction Based on Wavelet Analysis

Ni Lihua, Chen Xiaorong, Huang Qian
College of Computer Science and Information
Guizhou University
Guiyang, China
nilihua8707@126.com, xrchengz@163.com

***Abstract*—As the traffic flow has the features of nonlinear and strong interference, it has different characteristics in different time-frequency spaces. Firstly, this article uses the wavelet analysis method, decomposes a group of original traffic flow signals containing summarized information into series of time sequence signals that have different characters, then makes use of good linear fitting ability of the ARIMA model processes the wavelet analysis time signal through the ARIMA model. Using matlab and SPSS, the measured traffic flow data were analyzed verified. Experiment results show that the way of combining the wavelet analysis with ARIMA model can reduce the prediction error effectively, and improve the forecasting accuracy by about 80%, this way has high feasibility.**

Keywords- Wavelet analysis; ARIMA; traffic flow; traffic flow forecasting

I. INTRODUCTION

Traffic flow forecasting is the basic of traffic engineering、transport planning and traffic control, the traffic flow of timely and accurate forecasts is to improve the traffic flow distribution and control key factors, so the real-time traffic flow forecasting is becoming increasingly important. Currently, there are a lot of forecasting methods, but research shows that various methods have their advantages and disadvantages, no anyone one can meet predicting all of the time series. Moving Average (MA) is simple, but the forecast accuracy is not high; ARIMA, Kalman filtering and Bayesian network model have high forecast accuracy and mature technique, but their modeling methods are tedious; artificial intelligence and neural networks are fit for nonlinear fitting, but easily fall into local minimum, which have slow convergence; SVM model is adequate for predicting small samples, which is able to get the global optimum, but the forecast is very sensitive to the kernel function [1-2].

Bates, etc. having proposed the ideas of combination forecast in 1969, this method is like that, for some methods which participated in combination, choosing appropriate method to combine their results, to play their respective advantages and improve the prediction accuracy [3]. Wavelet analysis decomposed a set of original traffic signal which had comprehensive information into multiple time series with different signal characteristics, to reflect and distinguish the inherent trend and random perturbations between traffic signals; and ARIMA model had good ability of the linear fit. So this paper presents a hybrid model based on ARIMA and wavelet analysis (WAARIMA). First of all, taking the original traffic signal into wavelet decomposition and reconstruction, and then using ARIMA model to forecast traffic flow, and realizing analysis and forecasting with Matlab and SPSS. Combining the statistical methods with wavelet analysis, that gave full play to the advantages of both models to improve the prediction accuracy.

II. WAVELET DECOMPOSITION AND RECONSTRUCTION

Realizing wavelet decomposition by Mallat algorithm, Mallat is like that[4]:

$$\begin{cases} A_{j+1} = HA_j \\ D_{j+1} = GD_j \end{cases} \qquad (1)$$

In the formula, $j=0,1,\cdots,j-1$, $H, G,$ respectively stand for low pass and high pass filters. A_j is approximation coefficient after decomposition, D_j stand for detail coefficients.

Wavelet decomposition as shown in Fig. 1, defined A as the original signal X, so by the type X can be divided into $D_0, D_1, \cdots, D_j and A_j$ (*j* is the maximum decomposition level):

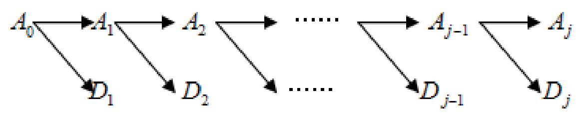

Figure 1. Wavelet decomposition process

Wavelet reconstruction process as shown in Fig. 2, the signal decomposed by Mallat algorithm can be reconstructed with the reconstruction algorithm, describing as:

$$A_j = HA_{j+1} + GD_{j+1}, j = j-1, j-2, \cdots, 0 \qquad (2)$$

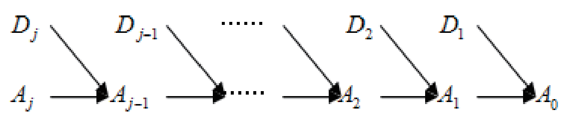

Figure 2. Wavelet reconstruction process

This research was supported by the Informatization Special Foundation Project of Guizhou Province of China in 2009. No. 0958.

978-1-4244- 7618-3 /10/$26.00 ©2010 IEEE

III. ARIMA MODEL

ARIMA model, also known as the famous Box-Jenkins model, which was a time series modeling, founded in the early 70s by Box and Jenkins, and it was a mixed form of AR and MA, which can be extended to analyze the time series including seasonal trends [5].

AR(p): $$v_t = \phi_1 v_{t-1} + \phi_2 v_{t-2} + \cdots + \phi_p v_{t-p} + \mu_t \quad (3)$$

MA(q): $$\mu_t = \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2} - \cdots - \theta_q \varepsilon_{t-q} \quad (4)$$

ARMA(p, q):

$$v_t = \phi_1 v_{t-1} + \phi_2 v_{t-2} + \cdots + \phi_p v_{t-p} + \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2} - \cdots - \theta_q \varepsilon_{t-q} \quad (5)$$

If taking a non-stationary time series by d time's difference, which becomes stable, then with a smooth ARMA (p, q) as its generation model, well then we say that the original time series as autoregressive integrated moving average time series, denoted as ARIMA*(p,d,q)*. Where AR is the autoregressive, *p* is the factor of autoregressive, MA is the moving average, *q* is the moving average cycle life, d is the difference frequency since time series become stationary[2], then the prediction model can be written as (5), In the formula, v_t are sample values, $\phi_i (i = 1,2,\cdots,p)$ and $\theta_j (j = 1,2,\cdots,q)$ are model parameters, ε_t is white noise sequence complied with normally distributed, *p,d,q* are the orders of model.

Generally, taking autocorrelation function (ACF) and partial autocorrelation function (PACF) to judge the time series' stationary, then determine the value of *p, q* [6], TABLE I.

TABLE I. THE ACF AND PACF THEORY OF ARMA (*P,Q*)

Model	ACF	PACF
AR(p)	Trailing; Decay to zero (Index or Sine attenuation)	truncation after p
MA(q)	truncation after q	Trailing; Decay to zero (Index or Sine attenuation)
ARMA(p,q)	Decay to zero (Index or Sine attenuation) after q	Decay to zero (Index or Sine attenuation) after p

If autocorrelation coefficient (ACF) increases with the lag, and rapidly tends to 0, so the time series is stationary.

If autocorrelation coefficient (ACF) increases with the lag, doesn't tend to 0, so the time series is not stationary. At the same time, there is a certain tendency in increase or decrease, so needing to put difference processing on data. If existing heteroscedasticity, what we need to do is taking technical processing on data, until the processed data from the correlation function and the partial correlation function value not significantly different from zero.

ARIMA modeling and forecasting have 5 steps:

step1: data validation. Based on time series' histograms、autocorrelation and partial autocorrelation function plot to test of its variance、trend and seasonal variation by ADF unit root, to identify stationary sequence.

step2: Stationary processing. Using the ACF and PACF processed non-stationary sequence to stationary.

step3: Model recognition. By analyze the ACF and PACF, and then through the AIC criterion (or SBC criteria) to determine the model's p, d, q values.

step4: Parameter estimation and model diagnostics. Using maximum likelihood estimation to obtain all of the estimated value on parameters, and tested in the model, diagnosis whether the residual series error is white noise or not to determine the appropriateness of the model, if it doesn't appropriate to re-estimate parameters.

step5: Using the model with appropriate parameters to predict. This article uses the ARIMA model realize modeling, which is the time series analysis module in the statistical analysis software package SPSS features to of the whole modeling process.

IV. TRAFFIC FLOW FORECASTING FRAMEWORK

A. Forecasting framework

ARIMA Model for Traffic Flow Prediction based on Wavelet analysis, the framework as shown in Fig. 3, $V^0(k)$ stands for the original time series signal in k time (superscript is measure). WD: Wavelet decomposition, IR: image reconstruction, SYN: synthesize.

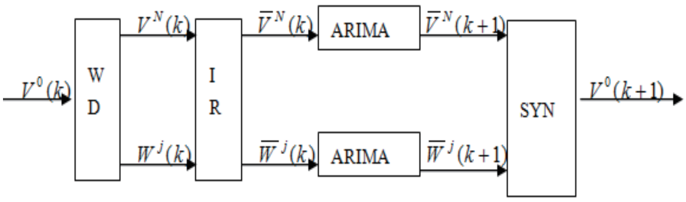

Figure 3. Hybrid model prediction framework based on wavelet analysis

The principle and steps as:

step1: Making use of appropriate wave functions break down the original traffic flow $V^0(k)$ to N level, the signal is decomposed into a set of approximate coefficients $V^N(k)$ and N group interference factor: $W^j(k), (j = 1,2,\cdots,N)$.

step2: Reconstructing the N+1 Groups of wavelet coefficients, getting (N+1) group of reconstructed time series signals: $\overline{V}^N(k)$ and $\overline{W}^j(k), (j = 1,2,\cdots,N)$.

Step3: Finally, adopting appropriate ARIMA model for each of the (N+1) groups of time series. Gaining (N+1) predicted values: $\overline{V}^N(k+1)$ and $\overline{W}^j(k+1), (j = 1,2,\cdots,N)$.

Synthesis of these forecasts, the results is prediction on short-term traffic flow.

$$V^0(k+1) = \overline{V}^N(k+1) + \sum_{j=1}^{N} \overline{W}^j(k+1), j = 1,2,\cdots,N \quad (6)$$

B. Predicted Performance

In order to select the appropriate forecasting method, requires appropriate performance indicators to measure the quality of methods. Assumed the original data is y_t, prediction is $\hat{y}_t$. Four measures as follows [1-2].

1) Mean Absolute Error

$$MAE = \frac{1}{n}\sum_{t=1}^{n} | y_t - \hat{y}_t | \quad (7)$$

2) Mean Square Error

$$MSE = \sqrt{\frac{\sum_{t=1}^{n}(y_t - \hat{y}_t)^2}{n}} \quad (8)$$

3) Mean Percent Absolute Error

$$MAPE = \frac{1}{n}\sum_{t=1}^{n}\left|\frac{y_t - \hat{y}_t}{y_t}\right| \times 100\% \quad (9)$$

4) Equal Coefficient

$$EC = 1 - \frac{\sum_{t=1}^{n}(y_t - \hat{y}_t)^2}{\sum_{t=1}^{n}(y_t + \hat{y}_t)^2} \quad (10)$$

The four evaluation indicators are the key indicator to evaluate predictive effect in the traffic flow forecasting. MAE is the absolute average error between predicted and actual values; MSE is not only able to respond to the error size, but also describe the concentration and degree of dispersion about the error distribution, the greater the mean square error, more discrete the error sequence, the forecast results is worse; MAPE is a comprehensive indicator to evaluate the whole forecasting process; EC main responses whether the forecast curve follow the observed curve trend or not.

V. Traffic Flow Forecasting

A. Data sources

The research object in this article is the traffic flow going straight from Guiyang Railway Station to the new intersection, recorded for the intraday traffic flow from 7:30 to19:30 to 5 minutes intervals. This article predicts 1000 traffic flow data on April 2, 2009 to April 10.

B. Experimental results and analysis

1) Wavelet decomposition and reconstruction

This article compares the figure used db2, db4 and db5 wavelet to decompose and reconstruct the time sequence to four levels. Finding out that using db5 wavelet get the curve of the fourth layer details is smoother, so db5 is chosen as wavelet. The result as shown in Fig. 4, where d1~d4 are the details of the decomposition (perturbation signal) of 1 to 4 layer, a4 stands for the 4th layer approximate weight obtained after decomposition (approximate signal).

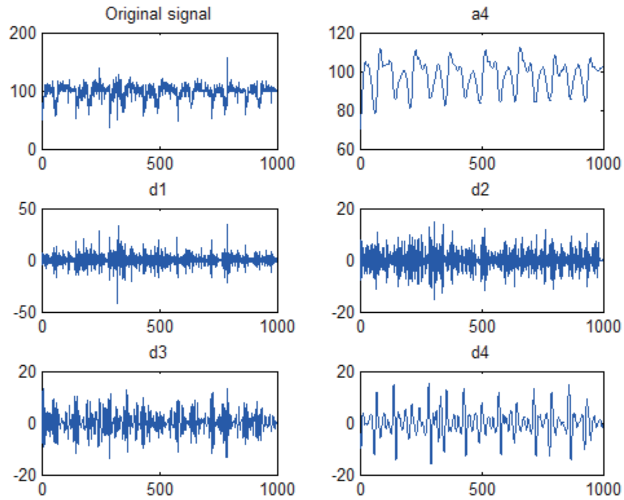

Figure 4. orignal and decomposition signal

2) ARIMA modeling and forecasting

Using ARIMA model making model for Low frequency signal and the high-frequency signals, the detail operations are is as follows:

Dealing with low-frequency signal a4 by the ARIMA model, and analysis of the autocorrelation function (ACF) and partial autocorrelation function (PACF), according to the AIC criterion, The model for a4 is ARIMA(5,1,5), obtained the predicted signal $\hat{a}4$, and then did the same operating for the high-frequency signals. The model for d1,d2,d3 and d4 is ARIMA(4,0,3), ARIMA(6,0,5), ARIMA(5,1,6), ARIMA(5,1,4) individually. So, gaining the predicted result-- $\hat{d}1, \hat{d}2, \hat{d}3$ and $\hat{d}4$, this paper uses ARIMA module in the SPSS to achieve this process. Establishing the value of p, d, q through model recognition, but also need to estimate the parameters, to determine whether the residuals are white noise or not, so as to determine the model's appropriateness, this article take advantage of the residual error autocorrelation function (ACF) to determine, in the ACF, if residual sequence falls into the random interval, then determine the model is appropriate.

3) Synthesis of the final prediction

If the predicted value is a2(t+1), then

$$a2(t+1) = \hat{a}2 + \hat{d}1 + \hat{d}2 + \hat{d}_3 + \hat{d}_4 \quad (11)$$

This article predicts 1000 traffic flow data on April 2, 2009 to April 10, and compared the predicted effect which separately obtained by ARIMA model (Fig. 5) and combination of ARIMA and wavelet prediction model (Fig. 6). As can be seen from the figure, the combined predicted value is better than ARIMA models.

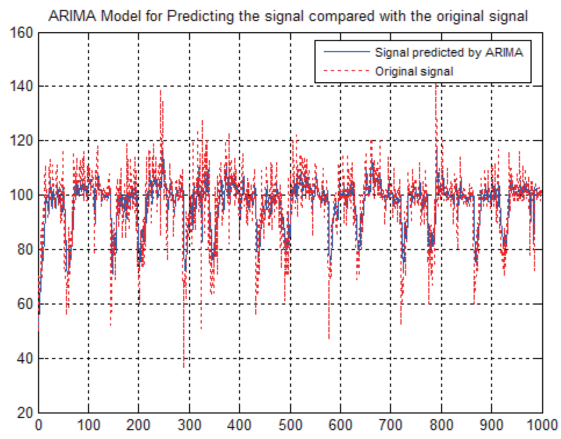

Figure 5. Predicted by ARIMA model

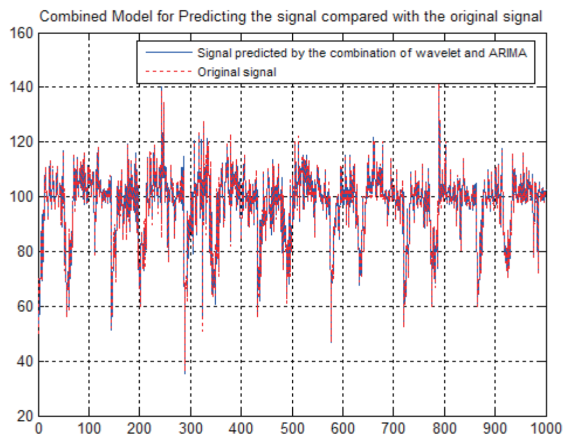

Figure 6. Predicted by ARIMA model and Wavelet

Comparison of the forecast performance: TABLE II.

TABLE II. FORECAST PERFORMANCE

measure	ARIMA model	Combined with wavelet and ARIMA
MAE	7.1969	1.1797
MSE	10.0939	1.6688
MPAE(%)	8.0150	1.2683
EC	0.9973	0.9999

Two kinds of forecasting models can be compared in MAE, MSE, MAPE and EC in the table. As can be seen from the comparison, the forecasting model after wavelet analysis has higher forecasting precision and lower error in comparison with those not being analyzed. EC mainly reflects whether forecasting curve follow the tendency of observing curve. EC after wavelet analysis is a few higher from the table. The predicted value of model being improved more effectively follows a tendency of observing curves. Competition between predicted value and actual value (Ten sets of data can be selected to show in TABLE III for lack of space):

TABLE III. COMPARED WITH PREDICTED VALUE AND ACTUAL VALUE

Actual value	ARIMA		Combined with wavelet and ARIMA	
	Predicted value	error	Predicted value	error
98	100.8163	-2.8163	99.1604	-1.1604
105	99.9223	5.0777	104.8867	0.1133
98	101.6663	-3.6663	98.0746	-0.07460
88	82.7732	6.1700	87.0221	1.2431
72	88.0065	-16.0065	72.9597	-0.9597
100	102.1488	-2.1488	99.8626	0.1374
102	102.6909	0.6909	101.6789	0.3211
103	101.8536	1.1464	103.8512	0.8512
101	99.87134	1.1287	101.0560	0.0560
98	102.6873	-4.6873	98.3753	0.3753

VI. CONCLUSION

As the traffic flow has the features of nonlinear and strong interference, this paper proposed ARIMA Model for Traffic Flow Prediction Based on Wavelet analysis. Because traffic flow in different scale space has different frequency characteristics, this article analyzes traffic flow through signal decomposition and reconstruction, and deals with the data after wavelet transform. This paper full uses of the "mathematical microscope" feature of wavelet transform. The nonlinear system of time series is decomposed into multiple components, each signal component dealt with ARIMA forecasts respectively, and finally back into the final prediction. This method is effective to improve the prediction accuracy. Experimental results show that this method can predict the traffic flow dynamically.

ACKNOWLEDGMENT

This research was supported by the Informatization Special Foundation Project of Guizhou Province of China in 2009. No. 0958. The authors are very grateful to the traffic police and the college of science teachers, for their constructive comments and suggestions, which have been very helpful in improving the paper.

REFERENCES

[1] Tan Manchun. A Hybrid ARIMA and SVM Model for Traffic Flow Prediction Based on Wavelet Denoising [J]. Jinan University, Guangdong, 2009, 7: 1~2 (In Chinese).

[2] Jiang Songtao. The study of method in short term traffic flow forecasting and empirical research [D]. Zhejiang University of technology, Zhejiang, 2009,11: 9~12 (In Chinese)

[3] BATES J M ,GRAN GER C W J.The Combination of forecasts [J]. Operational Research Quarterly, 1969, 20 (1): 451~468.

[4] Mallat S G. A theory for multi2resolution signaldecomposition :the wavelet representation [J]. IEEETransactions on Pattern Analysis and MachineIntelligence , 1989 , 11 (7) : 674~693.

[5] Wang Baobin. Network traffic prediction based on ARIMA model [J]. Handan College, Hebei 2008, 9(3): 31~32 (In Chinese).

[6] Niu Yijun. Application of ARIMA model in semiconductor demand forecasting [J]. University of Shanghai for Science and Technology, Shanghai, 2007, 5: 1~2 (In Chinese).

[7] Li hongmei. Prediction of the short traffic flow and visuliazation based matlab [D]. Shandong University of Science and Technology, Shangdong, 2007, 5: 11~12 (In Chinese).

[8] Tan Manchun. Traffic Flow Prediction Based on Hybrid ARIMA and ANN Model [J]. Jinan University, Guangdong, 2007, 7 (In Chinese).

[9] GAO Yong. Wavelet analysis2based NPR prediction of short-term traffic flow [J]. University of Science and Technology of China, An hui, 2008, 12. 1428~1429.

[10] Li Chunjun. Traffic Prediction Based on Wavelet Analysis [J]. School of Computer and Communication Engineering, Sichuan, 2003, 12. 7~8